

 意优科技		RP Series Specification Parameters					
No.		1	2	3	4	5	6
Module Model		RP40C	RP50L	RP50H	RP70L	RP70H	RP90L
Structure	Reducer Type	Planetary	Planetary	Planetary	Planetary	Planetary	Planetary
	Reduction Ratio	25.0000	19.5278	19.5278	20.0000	20.0000	21.9130
	Rated Torque (N·m)	2	5	10	12	17	35
	Peak Torque (N·m)	6	15	30	36	60	120
	Rated Joint Power (W)	25	79	157	163	223	421
	Backlash (Arcmin)	≤12	≤12	≤12	≤12	≤12	≤12
	Rated Output Speed (rpm)	120	150	150	130	125	115
	Peak Output Speed (rpm)	145	180	180	160	150	135
	Dimension (OD×Axial)(mm)	Φ45×52	Φ59×53.5	Φ59×59.5	Φ77×52.5	Φ77×58.3	Φ96.5×56
	Weight (g)(excl. cable)	225.00	410.00	465.00	670.00	760.00	1075.00
	Rated Joint Efficiency	70%	70%	78%	78%	78%	82%
	Hollow Shaft Diameter (mm)	4	5	5	6.0	6.0	8
	Housing Appearance (Color)	Matte Black Standard	Matte Black Standard	Matte Black Standard	Matte Black Standard	Matte Black Standard	Matte Black Standard
	Moment of Inertia (kg·mm ²)	8.18	26.15	28.66	72.97	93.21	269.40
	Radial Dynamic/Static Load Rating (N)	1170/980	2140/2740	2140/2740	2430/3490	2430/3490	2810/4680
	Axial Dynamic/Static Load Rating (N)	/	5100/18700	5100/18700	5500/2270	5500/2270	6440/30480
Hardware	Operating Voltage (VDC)	48±10%					
	Max Withstand Voltage (VDC)	90					
	Driver	Built-in					
	Bus Current (Arms)	1.2	2.6	4.5	5.5	7.5	12.5
	Encoder Resolution (bit)	Dual-channel: 17(input),19(output)					
	Driver Interface	Power (DC in DC out), CAN FD (In,Out)					
Software	Communication Protocol	CANOpen / CANFD (Custom Development Available)					
	Control Mode	Current, Velocity, Position, MIT Control Mode					
Reliability & Service Life	Ambient Operating Temp. (°C)	-10~45					
	Max Housing Temp Rise (°C)	≤60					
	Noise Level (dB) @1m	≤60					
	IP Rating	IP54					
	Service Life (h)	> 3000					